



‘inherently sensing textiles’

Chris Gilbey

Our Team



Our Customers Make Textiles

We make them smart...

N0BODY DENIM

- *"Our passion is perfect fit jeans. We believe that the new world of wearable devices shouldn't compromise that."*
- *"Imagine blew us away with their approach and concepts showing that the fibre itself can be a device."*

*John Coldilis
Managing Director*



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Out Of The Lab And Into The Market

- The first field trials of sensing “geotextiles” using graphene
- Anticipated to cover 2 million square meters* within 3 years
- The first product will require > 15 tonnes of graphene
- Solves a major environmental challenge – reduces risk from CSG



Civil Engineering

Real time detection/sensing: water leaks, stress, erosion...



Size And Scale



The Size Of The Projects



Fracking



Chemicals are injected into the earth to fracture rocks & release gas

Chemicals & waste saline water at elevated temperatures are pumped into ponds

Ponds are several hectares in size & require lining systems engineered to minimise groundwater leakage

Fracking



29 Hectares

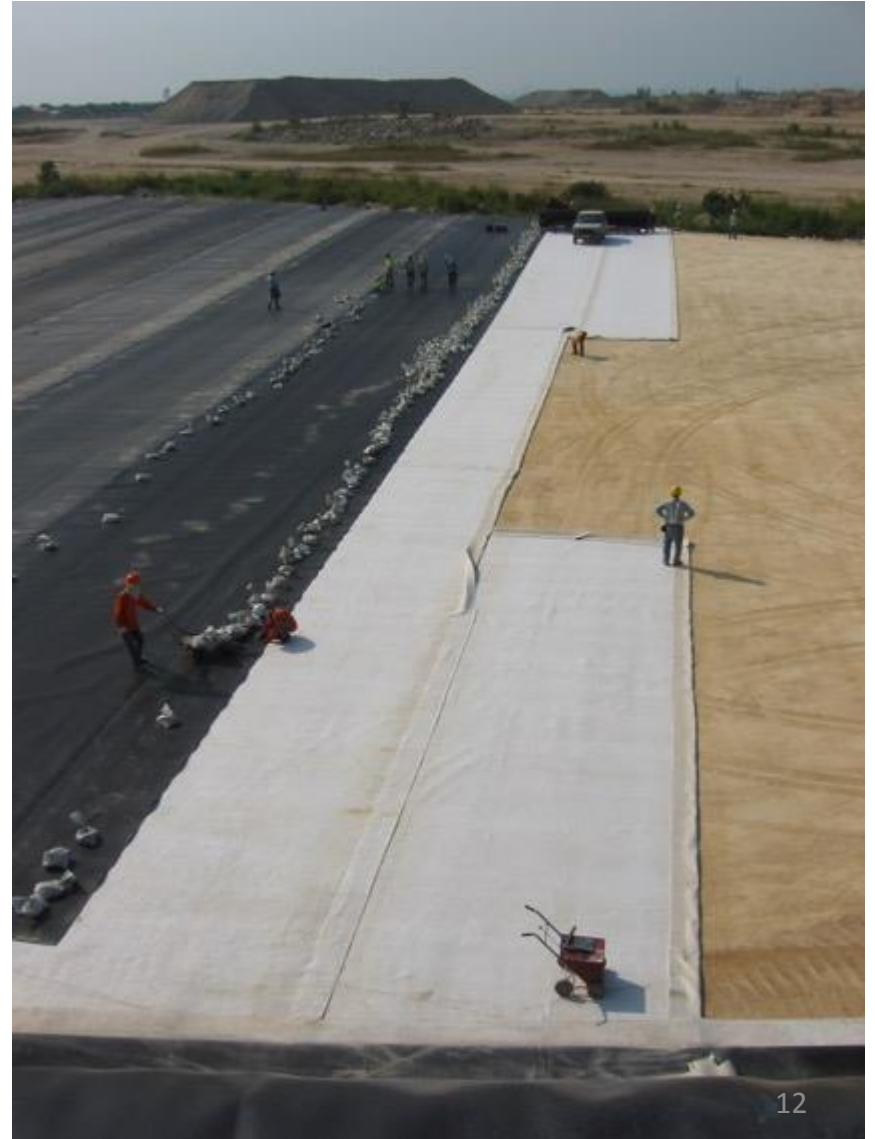


2074 Holes

Landfills



We deliver high quality leak detection without the need for a highly skilled workforce to install... saves time and money.



The Challenge - Continuous Conductivity Over Massive Areas

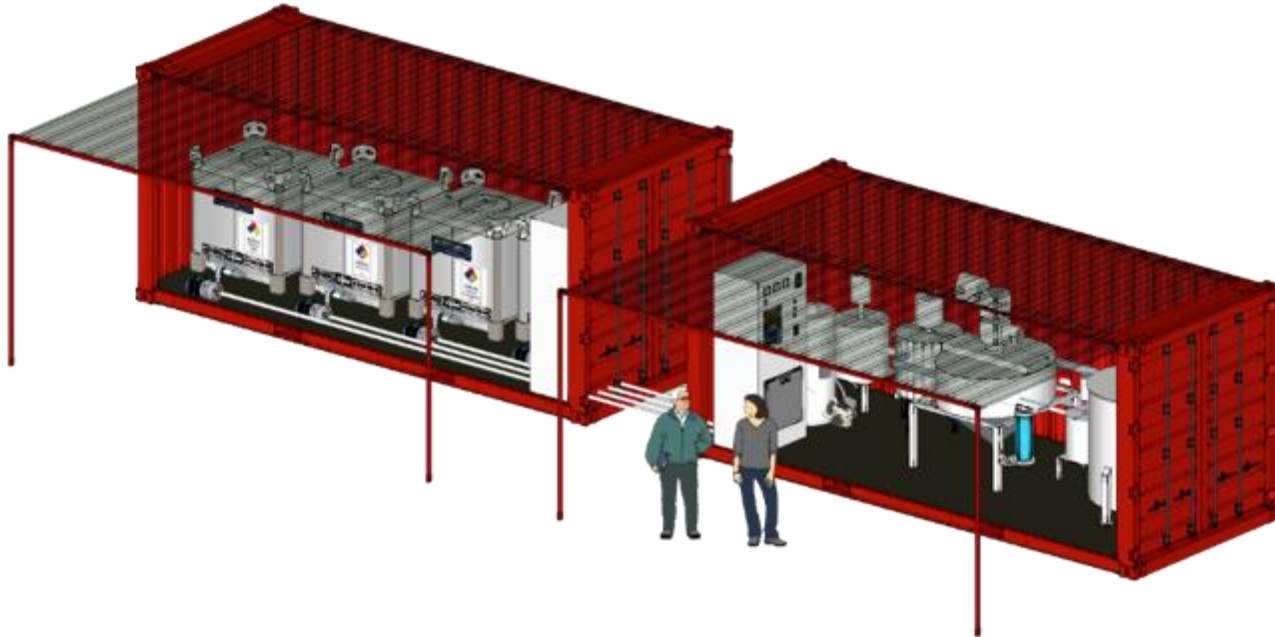


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Spark Testing



Functionalizing Graphene



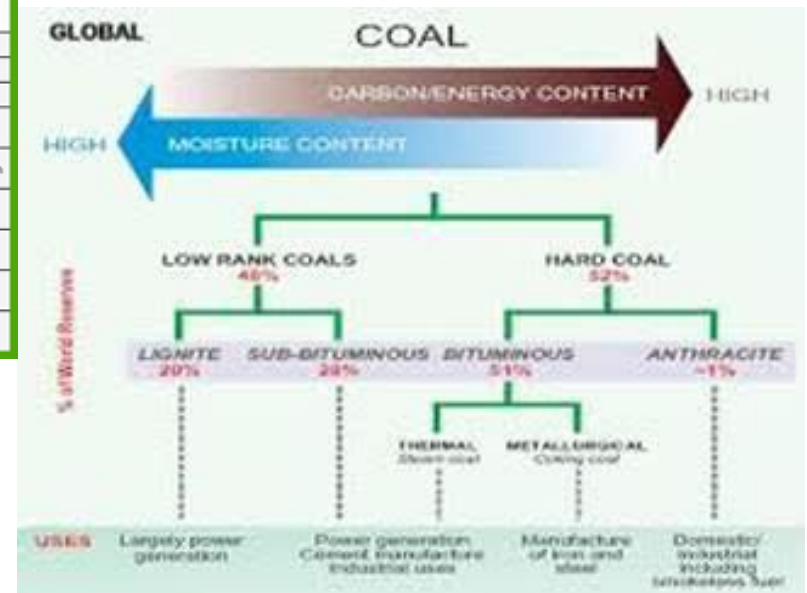
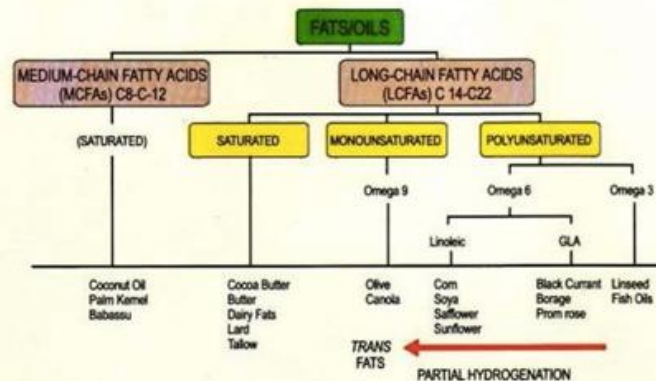
We are graphene agnostic.

We functionalize graphene to meet requirements of manufacturer, just in time.

Polymers, Coal, Oil, Steel... Graphene

	Polyethylene Terephthalate (PET)	High Density Polyethylene (HDPE)	Polyvinyl Chloride (PVC)	Low Density Polyethylene (LDPE)	Polypropylene (PP)	Polystyrene (PS)
Plastic ID Code	 PETE	 HDPE	 V	 LDPE	 PP	 PS
Clarity	Clear	Translucent	Clear	Translucent	Translucent	Clear
Moisture Barrier	Fair to Good	Good to Excellent	Fair	Good	Good to Excellent	Poor to Fair
Oxygen Barrier	Good	Poor	Good	Poor	Poor	Fair
Max. Temperature	120F	145F	140F	120F	165F	150F
Rigidity (Stiffness)	Moderate to High	Moderate	Moderate to High	Low	Moderate to High	Moderate to High
Resistance to Impact	Good to Excellent	Good to Excellent	Fair to Good	Excellent	Poor to Good	Poor to Good
Resistance to Heat	Poor to fair	Good	Poor to Fair	Fair	Good	Fair
Resistance to Cold	Good	Excellent	Fair	Excellent	Poor to Fair	Poor
Resistance to Sunlight	Good	Fair	Poor to Good	Fair	Fair	Poor to Fair

CLASSES OF FATS & OILS



Industries are known by a collective one name. The reality is that there are an infinite number of different subclasses of materials, each with vastly different qualities and functionality... Graphene is no different

We All Know That Graphene Needs To Be



Functional



Affordable



Scalable



Adaptive

You Can't Sell What You Can't Measure

Solution: Whole Of Supply Chain Certification



- Certification model reduces risk throughout supply chain
- We characterise and audit materials
- Ensure replicability and build trust...

Questions?